

Exercises 09/15

1. Show that if f is a contraction then it is continuous.
2. a. Let $f : (0, 1) \rightarrow \mathbb{R}$ be given by $f(x) = (1 - x)^{-1}$. What is the sup f ?
b. Let $f : (0, 1) \rightarrow \mathbb{R}$ be given by $f(x) = x$. What is the sup f ? Is it attained at a point in the set?
c. What happens when $(0, 1)$ is replaced by $[0, 1]$?
3. Let $f : [a, b] \rightarrow [a, b]$ be continuously differentiable. Suppose that $f'(x) \in (0, 1)$. Show that f is a contraction.
4. Let $y = f(k, l) = k^\alpha l^{1-\alpha}$ for $\alpha \in (0, 1)$. We may interpret y as output contingent on the inputs capital k and labor l . The functional form provided is called “Cobb-Douglas”. Compute $\frac{\partial y}{\partial l}$ and interpret. Compute the elasticities of output with respect to capital and labor. Compute the second partials of y and interpret.
5. Consider the Keynesian cross model of output determination, as given by

$$Y = C(Y - T) + I + G,$$

where Y is real gdp, T is lump-sum taxes, I is exogenous investment and G is government spending. Compute the government-spending multiplier, that is, let G change by dG and compute the corresponding change dY in Y . Hint: let

$$f(Y, G) = Y - C(Y - T) - I - G,$$

and compute the total differential df . Then, noting that for the equation to continue to hold after the change in G it must be that $df = 0$, set $df = 0$ and “solve” for dY in terms of dG .

6. Show that if $\dim \mathbf{W} = m$ and $\dim \mathbf{V} = n$ then $\dim \mathcal{L}(\mathbf{V}, \mathbf{W}) = nm$. Hint: $\mathcal{L}(\mathbf{V}, \mathbf{W}) \approx \mathbb{R}^{m \times n}$.